

Control work 2, and the quarter review

The whole of Chapter 1 in one paper, then an hour drawing the quarter together into a single map.

LESSON 26–27

2 × 40 MIN

ALGEBRA 10, NAZORAT ISHI 2

P1 · CHAPTERS 2–3 REVIEW

LESSON CLOCK

3

42'

10'

20'

10'

5'

Instructions	3 min
The paper	42 min
Answers on the board	10 min
Concept map in pairs	20 min
Targets for Quarter II	10 min
Homework	5 min

LEARNING OBJECTIVES

- Apply every idea of the quarter in one assessment.
- Choose the right tool without being told which section a question comes from.
- Build a concept map linking function, domain, range, property and transformation.
- Set a personal target for Quarter II.

TEXTBOOK

National	pp. 89–92
Cambridge	Pure Mathematics 1, pp. 56, 80
Workbook	Control paper B

01

Explanation

The paper — 30 marks, 42 minutes

Q	TASK	MARKS
1	Domain and range of $f(x) = \frac{3}{x-2}$	3
2	$f = 2x - 1, g = x^2$: find $fg(x), gf(x)$ and solve $fg(x) = gf(x)$	5
3	Find f^{-1} for $f(x) = \frac{x+1}{x-2}$ and state its domain	4
4	Test $x^4 - x^2$ and $x^3 + x$ for parity, showing $f(-x)$	4
5	Sign intervals of $f(x) = x^3 - 9x$	4
6	Describe the transformation from $y = x^2$ to $y = 2(x+1)^2 - 5$	4
7	A 48 m fence against a wall: find the greatest area, with the domain stated	6

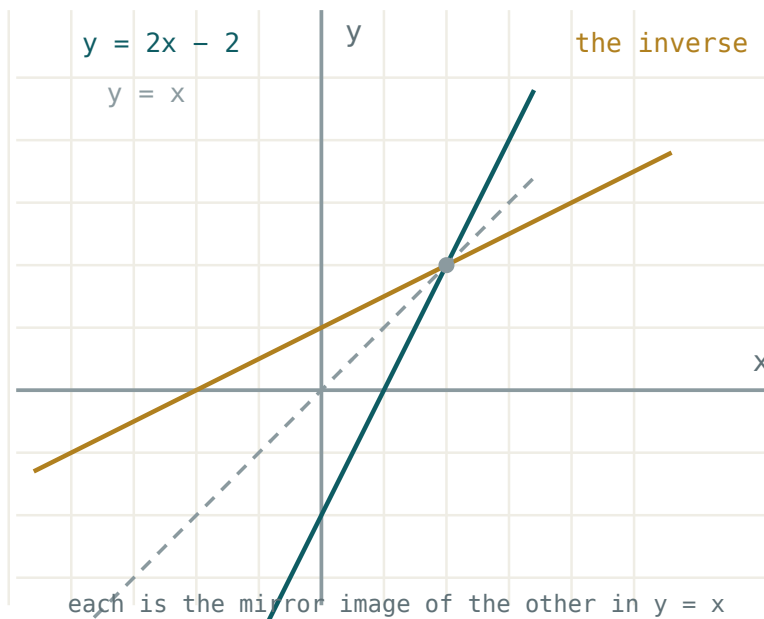
COMMAND WORDS

State — no working needed. **Show that** — every line must be there. **Find** — working plus answer.
Hence — you must use the previous part.

The concept map

In pairs, one A3 sheet. Six boxes, and every link labelled with a sentence:

- **mapping** → **function** — “exactly one output per input”
- **function** → **domain and range** — “what may go in, what comes out”
- **domain** → **one-to-one** — “restrict it and the function can be inverted”
- **one-to-one** → **inverse** — “reflect in $y = x$ ”
- **function** → **properties** — “zeros, sign, monotonic, parity, bounds”
- **function** → **transformations** — “inside the bracket works backwards”



The single picture that ties four boxes of the map together.

Two links must be the pair's own, not from this list.

Looking forward

Quarter II opens with rational and irrational equations. Everything in it depends on one habit built this quarter: **state the domain before you solve**. An irrational equation manufactures false roots precisely when the domain is ignored.

WRITE ONE TARGET

One sentence, in the exercise book, dated. Not “work harder” — something checkable, such as “state the domain in every question, in writing, before I start solving”.

02 Worked examples

EXAMPLE 1 Model answer, Q3: find f^{-1} for $f(x) = \frac{x+1}{x-2}$.

① $y(x-2) = x+1$

Clear the fraction.

② $yx - 2y = x + 1$

③ $x(y-1) = 2y+1$

Collect the x terms.

④ $x = \frac{2y+1}{y-1}$

Swap the letters.

ANSWER

$$f^{-1}(x) = \frac{2x+1}{x-1}, \text{ domain } x \neq 1$$

EXAMPLE 2 Model answer, Q7: 48 m of fence, three sides of a rectangle against a wall.

- ① Let each of the two equal sides be x m.

Third side $48 - 2x$.

- ② $A(x) = x(48 - 2x) = 48x - 2x^2$

Domain $0 < x < 24$.

- ③ $= -2(x - 12)^2 + 288$

- ④ $x = 12$, other side 24.

ANSWER

288 m^2 , a 12 m by 24 m pen

EXAMPLE 3 Model answer, Q5: sign intervals of $f(x) = x^3 - 9x$.

- ① $x(x - 3)(x + 3)$

- ② Zeros $-3, 0, 3$.

Four intervals.

- ③ Test $-4, -1, 1, 4$: $-28, 8, -8, 28$.

ANSWER

$f > 0$ on $(-3, 0) \cup (3, \infty)$; $f < 0$ on $(-\infty, -3) \cup (0, 3)$

03 Interactive model

Run through the six boxes of the map with the class before the pairs start.

The quarter in ten questionsDomain of $\sqrt{3x - 2}$:

$x \neq 2$

$x > 2$

$x \neq 3$

$\mathbb{R}$

Range of $\sqrt{3x - 2}$:

$y \neq 0$

$y \neq 2$

$y \neq 3$

$\mathbb{R}$

 $f = 2x - 1, g = x^2; fg(x)$ is:

$2x^2 - 1$

$(2x - 1)^2$

$4x^2 - 1$

$2x^2 + 1$

$x^4 - x^2$ is:

even

odd

neither

both

$x^3 + x$ is:

even

odd

neither

both

$y = 2(x + 1)^2 - 5$ is $y = x^2$:

left 1, stretch 2, down 5

right 1, stretch 2, down 5

left 1, stretch 2, up 5

left 2, stretch 1, down 5

The greatest area from 48 m against a wall is:

144 m^2

288 m^2

576 m^2

192 m^2

A function has an inverse exactly when it is:

The zeros of $x^3 - 9x$ are:

$f(x) = x^2$ is decreasing on:

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Control work	Nazorat ishi	Контрольная работа
Revision	Takrorlash	Повторение
Concept map	Tushunchalar xaritasi	Карта понятий
Summative assessment	Yakuniy baholash	Итоговое оценивание
Criterion	Mezon	Критерий
Target	Maqsad	Цель
Self-assessment	O'z-o'zini baholash	Самооценка
Command word	Topshiriq so'zi	Командное слово
Quarter	Chorak	Четверть

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

“Show that” means:

answer only

every line of working

a sketch

use a calculator

Before solving an equation with a fraction you should:

multiply out

state the domain

square both sides

guess

The vertex of $y = 2(x + 1)^2 - 5$ is:

$(1, -5)$

$(-1, -5)$

$(-1, 5)$

$(2, -5)$

A useful personal target is:

“work harder”

“be better at algebra”

“state the domain in writing every time”

“get 100%”

06

Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. Domain of $\frac{3}{x-2}$

ANSWER $x \neq 2$

2. Zeros of $x^3 - 9x$

ANSWER $0, \pm 3$

3. Is x^4 even or odd?

ANSWER even

4. Inverse of $f(x) = x + 9$

ANSWER $x - 9$

5. Vertex of $y = (x + 1)^2 - 5$

ANSWER $(-1, -5)$

6. $f = 2x, g = x + 1; fg(2)$

ANSWER 6

7. Range of $y = x^2 + 2$

ANSWER $y \geq 2$

Medium · the standard question

7 PROBLEMS

1. $f = 2x - 1, g = x^2$; solve $fg(x) = gf(x)$

ANSWER $x = 0$ or $x = 1$

2. Inverse of $\frac{x+1}{x-2}$

ANSWER $\frac{2x+1}{x-1}$

3. Sign intervals of $x^3 - 9x$

ANSWER > 0 on $(-3, 0) \cup (3, \infty)$

4. Describe $y = 2(x+1)^2 - 5$ from $y = x^2$

ANSWER left 1, stretch 2, down 5

5. 48 m against a wall — greatest area

ANSWER 288 m^2

6. Range of $\frac{3}{x-2}$

ANSWER $y \neq 0$

7. Parity of $x^3 + x$

ANSWER odd

Hard · stretch and reason

7 PROBLEMS

1. Domain of $\frac{\sqrt{x+2}}{x^2-9}$

ANSWER $x \geq -2, x \neq 3$

2. $f = \sqrt{x}, g = x - 4$; find fg and gf with domains

ANSWER $\sqrt{x-4}$ on $x \geq 4$; $\sqrt{x} - 4$ on $x \geq 0$

3. Restrict $f(x) = x^2 + 6x$ to invert it, then find f^{-1}

ANSWER $x \geq -3$; $f^{-1}(x) = -3 + \sqrt{x+9}$

4. Show that $f(x) = \frac{x}{x^2+1}$ is odd and bounded

ANSWER $f(-x) = -f(x)$; $|f| \leq 0.5$

5. 60 m of fence, two equal pens side by side against a wall. Maximise the total area

ANSWER $x = 10$, area 300 m^2

6. Find a, b with $y = a(x-2)^2 + b$ through $(0, 7)$ and $(2, -1)$

ANSWER $a = 2, b = -1$

7. Explain why f^{-1} has the domain of the range of f

ANSWER Inputs and outputs exchange roles

07 Homework

Homework — 4 tasks

Bring the concept map to the first lesson of Quarter II.

1. Rewrite every control-work question that lost a mark, in full.
2. Finish the concept map, including the two links your pair invented.
3. Write your target for Quarter II in one checkable sentence, and date it.

4. Domain and range of $f(x) = \sqrt{9 - x^2}$, with a sketch.